

DBMS Lecture 8: Keys & Candidate Key

Super Keys, Candidate Keys, Primary Keys & Minimality Rules

Topper's Revision Notes | Visual Diagrams, GATE Shortcuts & Interview Flashcards

1. What is a Key?

Before understanding Candidate Keys, we must first master the fundamental concept of a **Key**.

Definition (★)

A **Key** is an attribute or a set of attributes that can uniquely identify a tuple (row) in a relation (table).

In simple words: A key helps us uniquely identify each distinct record in a table so that individual tuples can be accessed, updated, or referenced without ambiguity.

2. Why Do We Need a Key?

Consider a `Student` table without any unique identification attribute:

Name	City	Age
Reddy	Bengaluru	20
Reddy	Bengaluru	20

This creates a critical problem for the database system:

- Are these two rows representing the **exact same student** mistakenly inserted twice?
- Or do they represent **two distinct students** who happen to share the exact same Name, City, and Age?

Using only `Name`, `City`, and `Age`, it is impossible for the DBMS to distinguish between these two tuples.

3. Adding a Unique Attribute

To resolve this ambiguity, we add an attribute specifically designed to guarantee uniqueness, such as `Roll_No`:

Roll_No	Name	City	Age
1	Reddy	Bengaluru	20
3	Reddy	Bengaluru	20

Now we can unambiguously identify each student:

- Roll_No = 1 $\rightarrow$ Student 1
- Roll_No = 3 $\rightarrow$ Student 2

Even though all other attribute values are completely identical, `Roll_No` distinguishes the two rows. Therefore, `Roll_No` acts as a valid **Key**.

4. Examples of Possible Keys

Depending on the real-world domain, a table may contain several potential unique identifiers:

Roll_No	Name	Aadhaar	Phone	Email
1	Reddy	XXXX-1234	9876543210	reddy@gmail.com
2	Ravi	YYYY-5678	8765432109	ravi@gmail.com

Possible key candidates in a student relation:

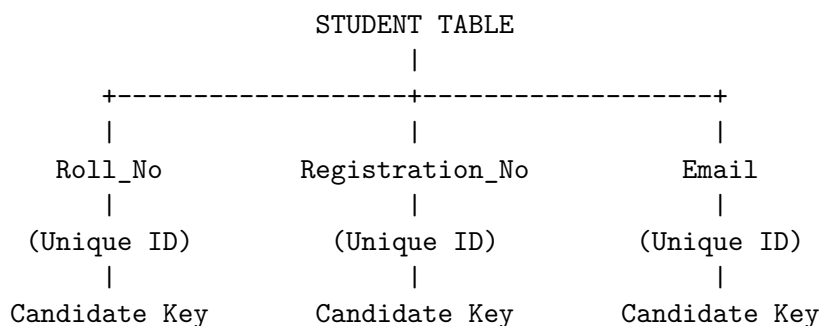
- Roll Number / Student ID / Registration Number
- Aadhaar Number / PAN Number / Voter ID / Passport Number
- Phone Number / Email ID

Important Real-World Caveat

Whether an attribute actually qualifies as a key depends strictly on the **business rules and integrity constraints** enforced by the organization, not just current sample data.

5. What is a Candidate Key? (★★★ High Frequency)

When a table has multiple attributes that can each independently identify a row, each of these valid options is called a **Candidate Key**.



Definition (★)

A **Candidate Key** is a **minimal super key** that can uniquely identify every tuple in a relation.

6. Why is "Minimal" Important?

Minimality is the single most critical aspect of the Candidate Key definition. A Candidate Key must satisfy two strict conditions:

1. **Uniqueness Property:** It must uniquely identify every row in the relation.
2. **Minimality Property:** No proper subset of attributes can be removed from it while still preserving uniqueness.

The Candidate Key Formula

Candidate Key = Unique + Minimal

7. Example of Candidate Keys

Consider the following Student dataset:

Roll_No	Registration_No	Email	Name	Age
101	REG001	a@gmail.com	Ali	20
102	REG002	b@gmail.com	Ravi	21
103	REG003	c@gmail.com	Sara	19

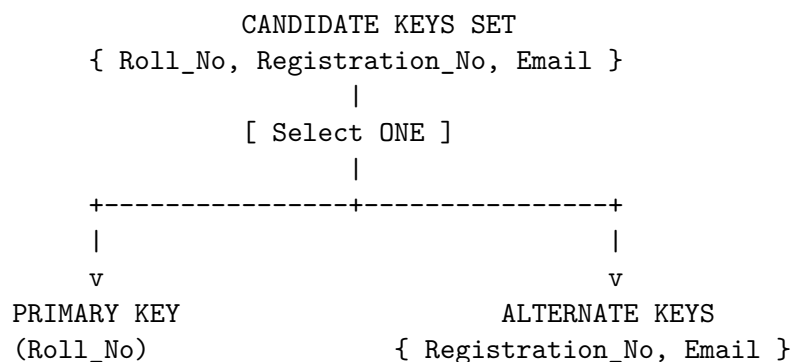
Assuming business constraints guarantee that Roll_No, Registration_No, and Email are all unique across all students:

Candidate Keys = {{Roll_No}, {Registration_No}, {Email}}

Each of these set choices independently and uniquely identifies a student record.

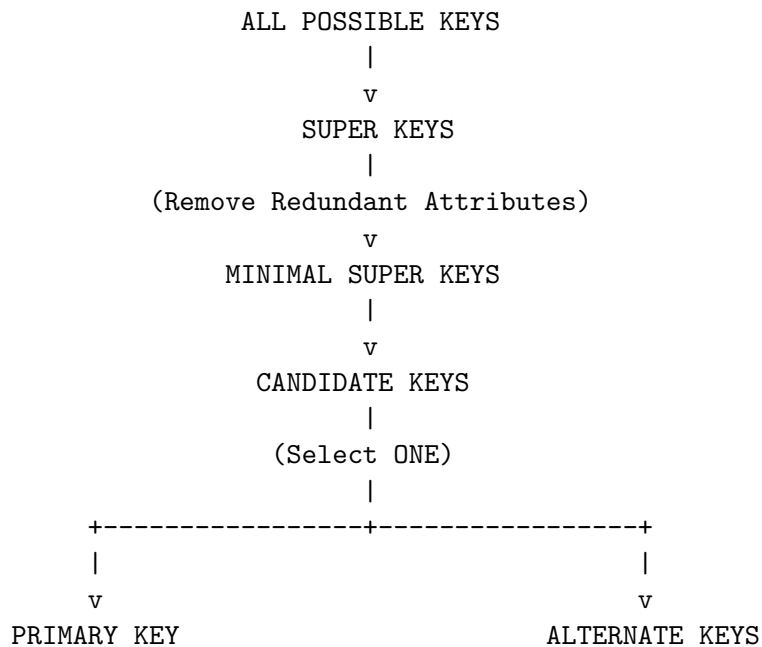
8. Choosing the Primary Key & Alternate Keys

Once all Candidate Keys are identified, the Database Administrator (DBA) selects **exactly one** candidate key to serve as the **Primary Key**.



- **Primary Key:** The chosen candidate key used as the main record identifier.
- **Alternate Keys:** The remaining candidate keys that were not selected as the primary key.

9. Candidate Key → Primary Key → Alternate Keys Hierarchy



10. Candidate Key vs. Primary Key

Dimension	Candidate Key	Primary Key
Quantity	A relation can have **multiple** candidate keys.	A relation can have **only one** primary key.
Selection	Represents all valid minimal candidate options.	Exactly one key chosen from candidate keys.
Status	Non-selected ones become **Alternate Keys** .	The primary identification mechanism.
Uniqueness	Must uniquely identify tuples.	Must uniquely identify tuples.
Minimality	Must strictly be minimal.	Inherits minimality from candidate key.

Table 1: Comparison: Candidate Key vs. Primary Key

11. Crucial Relationship (★★★ Exam Statement)

Core Rule to Remember

Every primary key is a candidate key, but every candidate key is not necessarily the primary key.

Reasoning: The primary key is chosen from the candidate key set. Thus, all primary keys belong to the candidate key set, but unselected candidate keys remain alternate keys.

12. Candidate Key vs. Super Key

This distinction is extremely critical for GATE and competitive exams:

- **Super Key:** Any set of attributes that uniquely identifies a tuple in a relation (can contain redundant attributes).
- **Candidate Key:** A minimal super key (contains zero redundant attributes).

Super Key vs Candidate Key Rule

Every Candidate Key is a Super Key, BUT Not every Super Key is a Candidate Key!

Detailed Example

Given schema `Student(Roll_No, Name, Age)` where `Roll_No` is unique:

- $\{Roll_No\} \rightarrow$ **Candidate Key** (Unique + Minimal).
- $\{Roll_No, Name\} \rightarrow$ **Super Key, NOT Candidate Key** (Unique, but `Name` is redundant).
- $\{Roll_No, Age\} \rightarrow$ **Super Key, NOT Candidate Key** (`Age` is redundant).

If we drop `Name` from $\{Roll_No, Name\}$, the remaining set $\{Roll_No\}$ still preserves full uniqueness. Hence, $\{Roll_No, Name\}$ fails the minimality test.

13. Composite Candidate Keys

A Candidate Key does not need to consist of only a single attribute. When single attributes cannot guarantee uniqueness individually, a combination of two or more attributes can form a key.

Definition

A **Composite Candidate Key** is a candidate key composed of two or more attributes acting together to uniquely identify a relation tuple.

Enrollment Example

Consider an Enrollment relation:

Student_ID	Course_ID	Semester	Grade
101	C01	1	A
101	C02	1	B
102	C01	1	A

- Student_ID alone is ****not unique**** (101 appears twice).
- Course_ID alone is ****not unique**** (C01 appears twice).
- However, the composite pair {Student_ID, Course_ID} uniquely identifies each enrollment record.

Therefore, {Student_ID, Course_ID} forms a ****Composite Candidate Key****.

14. Candidate Key Properties (★★★)

1. **Property 1 — Uniqueness:** No two distinct tuples in the relation can have identical values for the candidate key attributes.
2. **Property 2 — Minimality:** No proper subset of the candidate key attributes can uniquely identify the tuples on its own.
3. **Property 3 — Non-Nullity (In RDBMS implementations):** Attributes forming a candidate key meant to identify records should not accept NULL values.

15. Real-World Constraints & NULL Values

In academic theory, real-world attributes like Phone Numbers, Aadhaar, or Email Addresses are frequently used as examples of Candidate Keys. However, in production database engineering:

Production Engineering Realities

- **Phone Numbers:** May be shared by family members or left NULL.
- **Email Addresses:** Users may not possess an email or may change it.
- **Aadhaar / Passport:** Foreign citizens or minors may not have one.

A candidate key must strictly guarantee ****Uniqueness, Minimality, and Non-Nullability**** under all business conditions.

16. GATE Procedure: How to Find Candidate Keys

When solving GATE problems involving a relation R and a set of Functional Dependencies (FDs):

```
START GATE ALGORITHM
|
Step 1: Compute Attribute Closures (X+) for subsets.
|
Step 2: Identify attribute sets whose closure contains ALL attributes of R.
      (These are SUPER KEYS).
|
Step 3: Check each Super Key for MINIMALITY.
      (Ensure no subset of it can determine all attributes).
|
Step 4: The minimal super keys form the CANDIDATE KEYS set.
```

17. Comprehensive Key Terminology Summary

Key Type	Core Definition	Key Characteristics
Super Key	Any attribute set that uniquely identifies tuples.	Can contain redundant/extra attributes.
Candidate Key	A minimal super key.	Unique + Minimal (No redundancy).
Primary Key	The chosen candidate key.	Unique, Minimal, strictly NOT NULL.
Alternate Key	Unselected candidate keys.	Candidate keys not picked as primary key.
Composite Key	Key consisting of multiple attributes.	Requires ≥ 2 columns combined.
Foreign Key	References primary key of another table.	Maintains Referential Integrity.

Table 2: Comprehensive Key Terminology Quick Reference

18. Top 10 Interview & GATE Questions

Q1. What is a key in DBMS?

Answer: An attribute or set of attributes that uniquely identifies a tuple (row) in a relation.

Q2. What is a candidate key?

Answer: A minimal super key that uniquely identifies every tuple in a relation without redundant attributes.

Q3. What are the two essential properties of a candidate key?

Answer: Uniqueness and Minimality.

Q4. Can a table have multiple candidate keys?

Answer: Yes. A table can have multiple candidate keys (e.g., Roll_No, Registration_No, Email).

Q5. Can a table have multiple primary keys?

Answer: No. A table can have only **one** primary key, though that primary key may be composite (multi-attribute).

Q6. How is the primary key selected?

Answer: The primary key is chosen by the Database Administrator from the available set of candidate keys.

Q7. What are alternate keys?

Answer: Alternate keys are the candidate keys that were not chosen as the primary key.

Q8. Is every candidate key a super key?

Answer: Yes, because a candidate key uniquely identifies tuples.

Q9. Is every super key a candidate key?

Answer: No. A super key may contain redundant attributes that violate minimality.

Q10. What is a composite candidate key?

Answer: A candidate key that consists of two or more attributes working together to uniquely identify rows.

19. Rapid Revision & Ultimate Memory Formula

The Key Reduction Sequence

SUPER KEY $\xrightarrow{\text{Remove Redundant Attributes}}$ CANDIDATE KEY $\xrightarrow{\text{Select One}}$ PRIMARY KEY

Candidate Keys – {Primary Key} = **ALTERNATE KEYS**

One-Line Exam Definitions

- **Key:** Uniquely identifies a tuple.
- **Super Key:** Any unique identifier (may have extra columns).
- **Candidate Key:** Minimal super key (no extra columns).
- **Primary Key:** The selected candidate key.
- **Alternate Key:** Candidate key not selected as primary.
- **Composite Key:** Key consisting of multiple attributes.

DBMS Study Roadmap Progress

- Lecture 1: Database, DBMS & RDBMS
- Lecture 2: File System vs DBMS
- Lecture 3: 2-Tier & 3-Tier Architecture
- Lecture 4: Schema in Database
- Lecture 5: Three-Schema Architecture
- Lecture 6: Data Independence
- Lecture 7: Integrity Constraints
- **Lecture 8: Keys & Candidate Key (COMPLETED ✓)**
- *Next Topic: Primary Key, Foreign Key, Alternate Key, Composite Key & Finding Candidate Keys using Functional Dependencies (GATE Problems)*